

RiskPulse

AI-Powered Financial Risk Intelligence & Stress Testing

Code to Connect: S&P Global and Crisil | Hackathon 2026

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1. Problem & Approach

The Challenge:

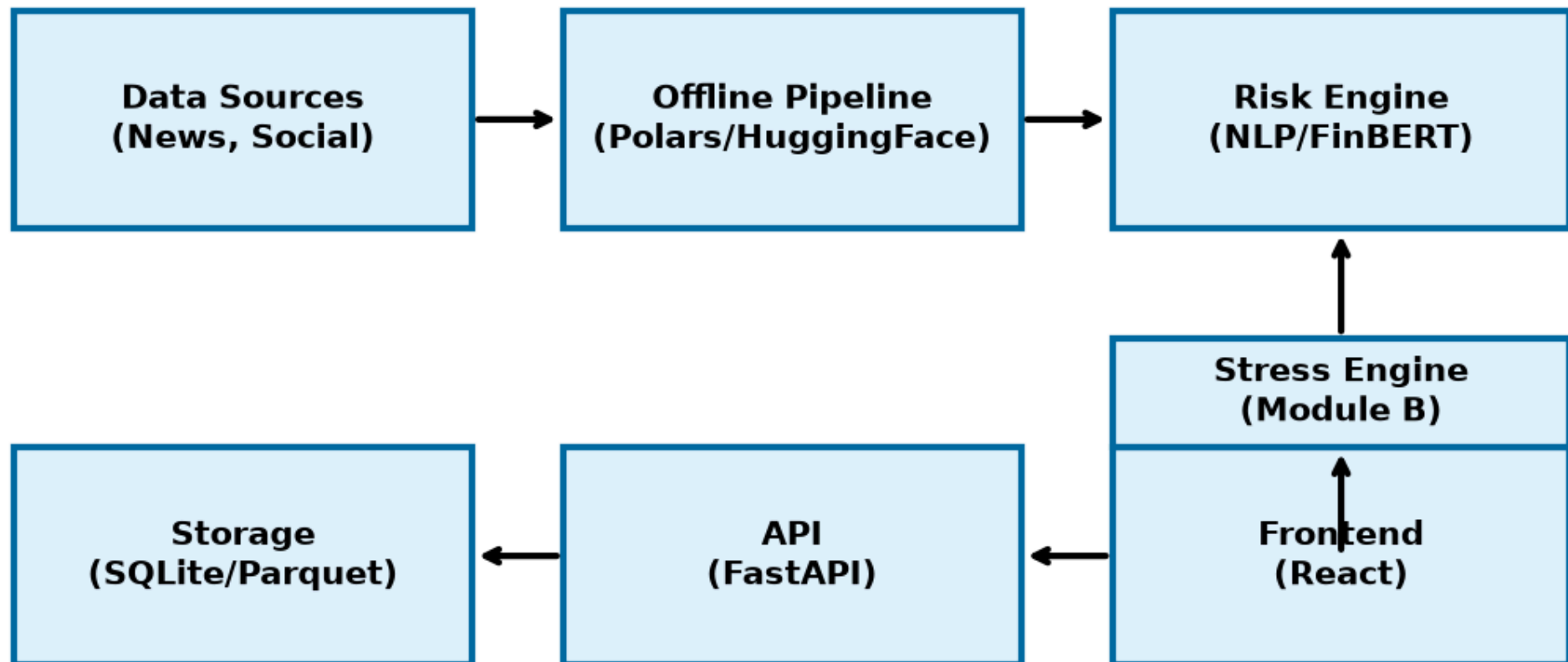
Financial risk emerges from unstructured, real-time events. Manually tracking news, predicting its impact, and calculating immediate portfolio exposures is too slow.

Our Approach:

- **Unified NLP Engine:** Ingests unstructured financial news to extract entities, sentiment, and categorize events.
- **Explainable Impact Scoring:** A hybrid deterministic rule-based formula that measures severity.
- **Automated Stress Testing (Module B):** High-impact events auto-trigger shocks against a synthetic wholesale-banking portfolio.
- **Dual-Mode Replay System:** A deterministic replay engine allows offline review of historical crises.

2. System Architecture

RiskPulse Architecture



3. NLP Deep Dive & Explainability

We prioritize accuracy and explainability:

- **Entity Resolution:** spaCy NER matched with RapidFuzz aliases mapping to the S&P 100 universe.
- **Sentiment:** FinBERT classifies financial tone (Pos, Neg, Neu).
- **Event Classification:** all-MiniLM embeddings compare news against a 12-class taxonomy, augmented with lexical rules.
- **Explainable Impact:**
 - $\text{Impact} = f(\text{Severity}, \text{Sentiment}, \text{Market Relevance})$
 - *Example output:* "Impact 8.4 — Geopolitical event +2.5; strong negative sentiment +1.8; high market relevance +1.7"

4. Module B: Portfolio Stress Testing

The Portfolio:

A synthetic \$7.3B wholesale-banking book containing Loans, Bonds, and Derivatives across major sectors.

The Trigger:

When the NLP engine scores an event as **High Severity** (Impact ≥ 7) and Adverse, a stress test triggers automatically.

The Execution:

Looks up predefined shocks (e.g. Geopolitical: Equities -12%, Rates -30bps) and recalculates the total portfolio MTM instantly.

5. Key Results & Validations

- **Dataset Used:** Evaluated on `zeroshot/twitter-financial-news-sentiment` and `topic` datasets, and integrated into the offline pipeline.
- **Performance:**
 - FinBERT Accuracy: ~66% (vs baseline mappings)
 - Inference Throughput: ~47 docs/sec (Sentiment), ~140 docs/sec (Events) on CPU.
- **End-to-End Reliability:** 100% deterministic Replay mode allows repeatable demonstrations without API rate limits.

6. Domain Impact & Next Steps

Domain Impact:

RiskPulse bridges the gap between qualitative news and quantitative risk management, providing risk officers with an actionable, instant view of idiosyncratic and macroeconomic shocks.

Next Steps:

- Introduce real-time streaming (Kafka/SSE).
- Calibrate shock priors using historical event-studies.
- Integrate Module A (Tactical Rebalancer) consuming the same generic Risk Signals.